

ARNAV BHATIA

Artificial Intelligence Machine Learning (AIML) Engineer | LLMs · Generative AI · MLOps

Delhi, India | +91-9582085559 | arnavbhatiamait@gmail.com | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building and deploying end-to-end intelligent systems, from LLM-powered RAG applications to production MLOps pipelines. Skilled in deep learning, generative AI, and cloud infrastructure (AWS, Docker, Terraform, CI/CD), with a track record of shipping cost-optimized, scalable systems.

TECHNICAL SKILLS

- Languages & Web:** Python, JavaScript, TypeScript, Next.js, React, FastAPI, Node.js, HTML5/CSS3, WebSockets
- AI/ML & Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, RAG, Agentic & Multi-Agent Systems, Prompt Engineering, GANs, Stable Diffusion, YOLO, OpenCV, AutoGen, Google ADK, OpenAI SDK, MCP
- Cloud & DevOps:** AWS, Google Cloud, Docker, Terraform, GitHub Actions, CI/CD, Kafka, Redis, Linux, Git
- Data & Analytics:** Pandas, NumPy, Scikit-learn, SQL (PostgreSQL), Tableau, Power BI

WORK EXPERIENCE

PAN Science Innovations | *LLM Engineer Intern* | Delhi, India

Aug 2025 – Present

- Co-developed [Nyaay AI](#), a legal RAG search platform with an AI chatbot that automates legal research and document workflows.
- Optimized [Council CRM](#), an AI-integrated platform, streamlining administrative workflows and client interaction tracking.
- Co-developed [OpticAll AI](#): built an asynchronous audio-processing pipeline with **AWS SQS, Kafka, and Redis** for message queuing, integrated **Gemini batch processing** for analytics, and deployed to **AWS ECS** via **GitHub Actions**.
- Built **The Benchmark Hub**, an LLM benchmarking platform evaluating model accuracy, latency, and domain robustness across multiple providers.

Aura AI | *Data Science Intern* | Remote

Jun 2024 – Sep 2024

- Built deepfake-detection models using custom CNN and XceptionNet architectures.
- Trained an SRGAN-based super-resolution model to upscale low-fidelity images by 4x.
- Built GenAI applications with Stable Diffusion and ControlNet, deployed via FastAPI and Streamlit.

PNB Housing Finance | *Salesforce Intern* | Noida, India

Oct 2024 – Nov 2024

- Optimized the Loan Management System (LMS), streamlining manual workflows and reducing processing latency.
- Built and integrated REST APIs with complex database queries to support backend workflows for mortgage finance systems.

PROJECTS

- Low-Cost Real-Time Voice Calling Agent** | [Demo Video](#): Built a full-duplex, low-latency voice-calling system (Starlette WebSockets + FastAPI) running at **~₹20 (~\$0.24)/hour**. Integrated Whisper/Groq ASR, multi-provider LLMs (Groq, OpenAI, Gemini, Ollama), and real-time TTS (Kokoro, Google Cloud, Indic Parler), with asyncio-based concurrency for instant user barge-in.
- Social Media Content Automation** | [GitHub](#) | [Demo](#): Built an automated content pipeline using **Gemini 2.5 Flash** (via LangChain), **FLUX.1-schnell**, **SDXL**, and Google Cloud TTS, with FFmpeg subtitle burning and GCS signed uploads. Automated twice-daily publishing to Instagram, Facebook, and YouTube via GitHub Actions.
- ESRGAN Image Super-Resolution** | [Hugging Face Space](#): Fine-tuned an ESRGAN architecture in PyTorch to upscale low-fidelity images 4x while preserving texture and structural detail; optimized for CPU/GPU efficiency.
- Multi-Architecture Landmark Recognition** | [Hugging Face Space](#): Built an ensemble vision model (ResNet50, EfficientNetB3, ViT) trained on 9k+ images for landmark recognition.

EDUCATION

- Guru Gobind Singh Indraprastha University** | B.Tech, Artificial Intelligence Machine Learning | **CGPA - 8.8** | **2022-2026**
- N.K. Bagrodia Public School** | Class X – **91%** (2019) | Class XII – **91%** (2021)

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) & AWS Certified AI Practitioner (AIF-C01)** | [CLF-C02](#) & [AIF-C01](#)
- Machine Learning Specialization** — Stanford University & DeepLearning.AI | [Credential](#)
- Deep Learning Specialization** — DeepLearning.AI | [Credential](#)
- Agentic AI & Multi-Agent Systems** — LangGraph, LangChain & Microsoft AutoGen | [AutoGen](#) & [LangGraph](#)
- Generative Adversarial Networks (GANs)** — DeepLearning.AI | [Credential](#)
- Complete Generative AI Course with LangChain & Hugging Face** — Krish Naik | [Credential](#)